



# Download files from the internet

Here, you can see the files that are being used to build up the assignment.

## Draw a house using line art

To generate a house, we put together the previous examples:

1. Build the roof using lines intersecting at a point.
2. Use rectangles to build the rest

## Assignment

### Generate frames using line ar

To make videos, we first need to setup a grid for all remaining video frames.

To do this, right after creating the video, we use the `prepVideo()` function to make all video frames be the same.

Here is an example use:

```
NiceGr = cuteGraph()  
NiceGr.prepVideo(minX=-5, minY=-5, maxX=10, maxY=10, magFactor=3)
```

Here, we tell the grid that we have that our house will be plotted approximately within the bounds of `(minX, minY)` to `(maxX, maxY)`. To make extra space for the names and the frontyard of the house, we make this frame three times larger using `magFactor=3`.

After we are done plotting each video frame, we save a picture using:

```
NiceGr.plotAll()  
NiceGr.saveImage("frame2.png")
```

We will collect these frames to make a video!

## Assignment

Feel free to experiment with different designs.

What can you draw?

```
In [2]: # Create the object
NiceGr = cuteGraph()
NiceGr.prepVideo(minX=-5, minY=-5, maxX=10, maxY=10, magFactor=3)

# Teach line above. Keep as one lesson. Fix the updates.

NiceGr.rect(x1=-5, y1=-10, x2=+5, y2=-5, color="Purple")

# Show the graph
NiceGr.plotAll()
NiceGr.saveImage("frame1.png")

# The door
NiceGr.setwidths(linewidth=3, pointwidth=1)
NiceGr.rect(x1=-1, y1=-10, x2=1, y2=-7, color="pink")

# The window
NiceGr.setwidths(linewidth=3, pointwidth=1)
NiceGr.rect(x1=+2, y1=-7, x2=2+1, y2=-7+1, color="silver")
NiceGr.rect(x1=-3, y1=-7, x2=-3+1, y2=-7+1, color="silver")
NiceGr.plotAll()
NiceGr.saveImage("frame2.png")

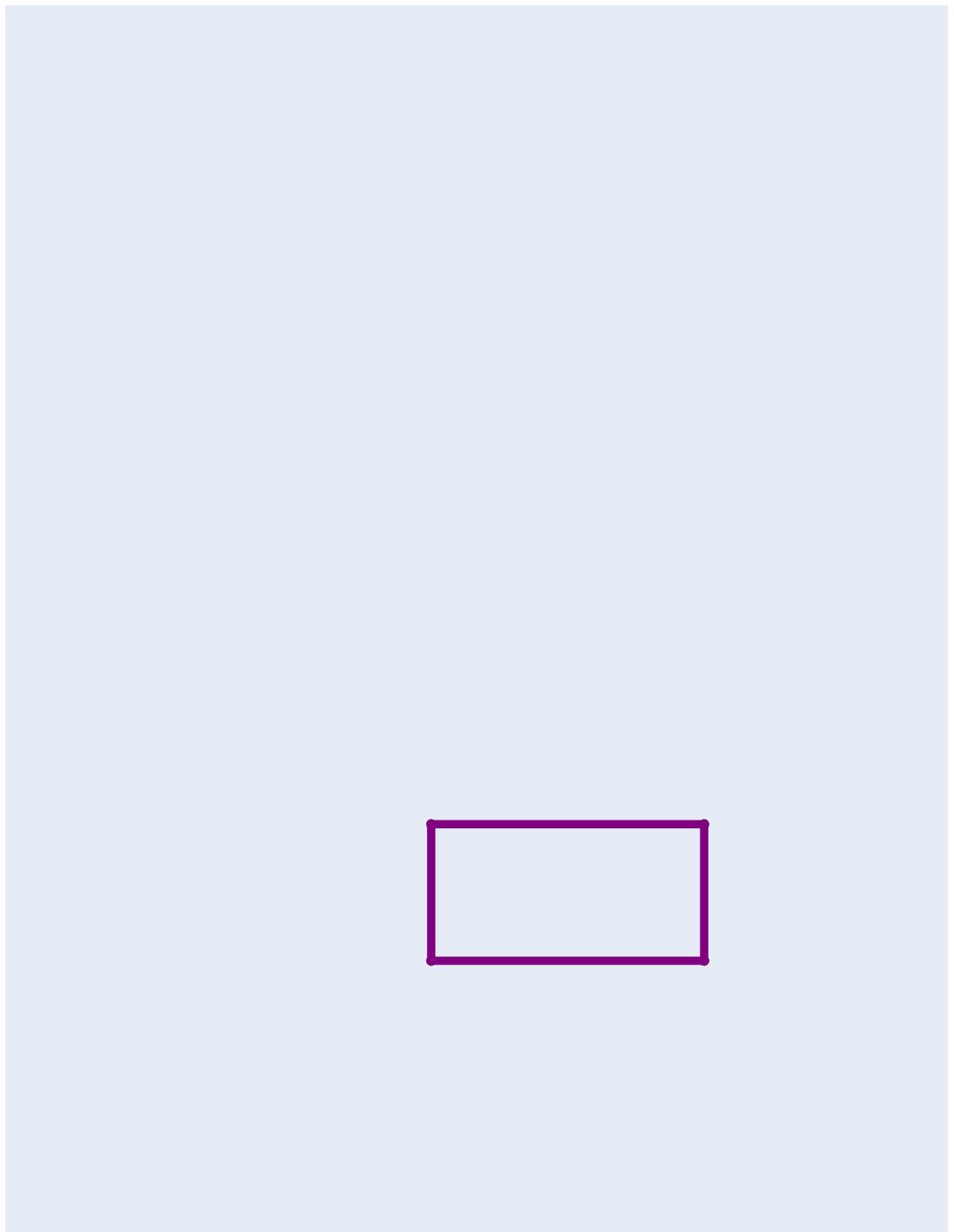
# The roof
for x1 in range(-5, +5+1):
    NiceGr.lineseg(x1=x1, y1=-5, x2=0, y2=0, color="light blue")

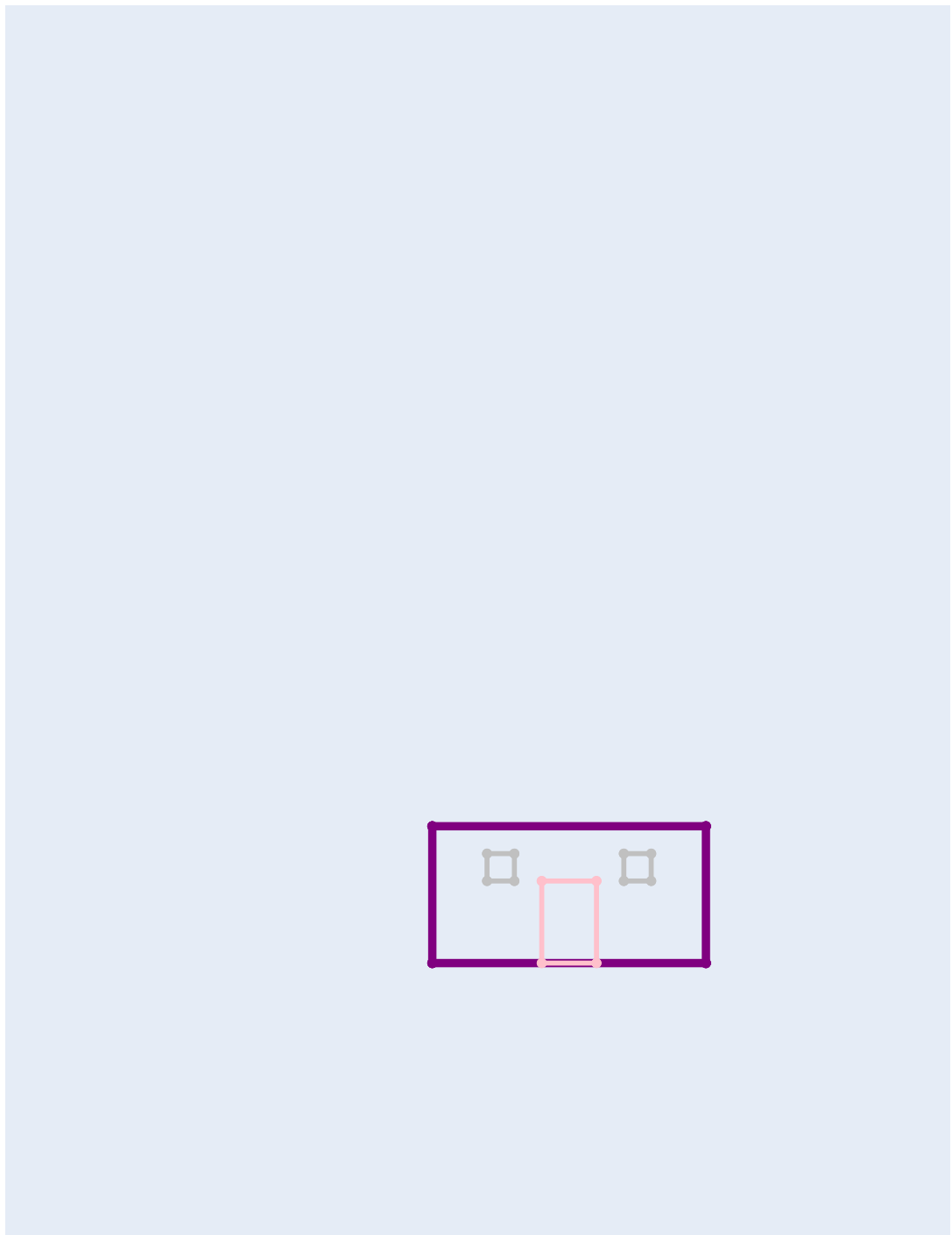
# Show the graph
NiceGr.plotAll()
NiceGr.saveImage("frame3.png")

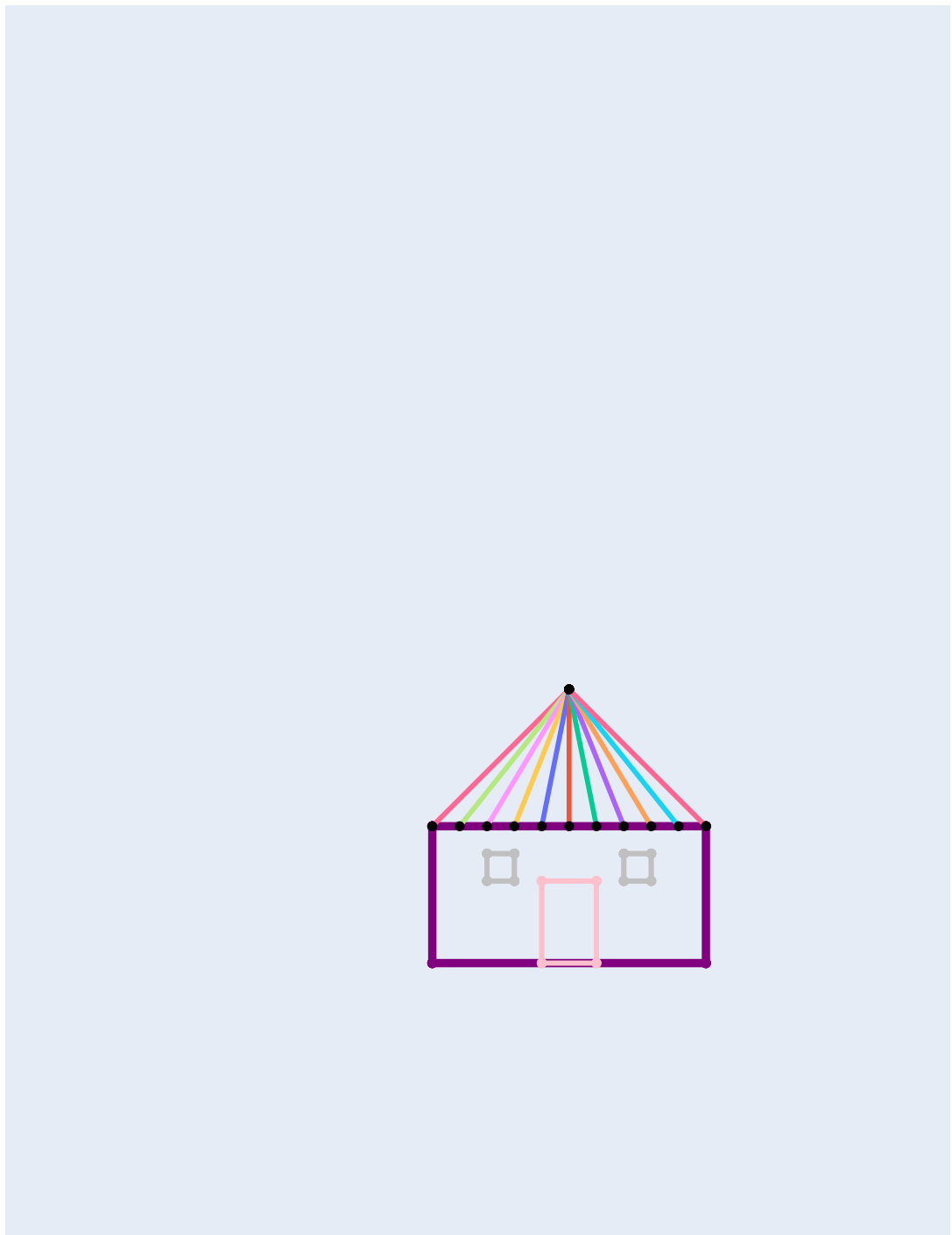
# Add a front:
y1 = -15
dx = 5
dy = 4
for x1 in range(-15, +10):
    x2 = x1 + dx
    y2 = y1 + dy
    NiceGr.lineseg(x1=x1, y1=y1, x2=x2, y2=y2, color="green")

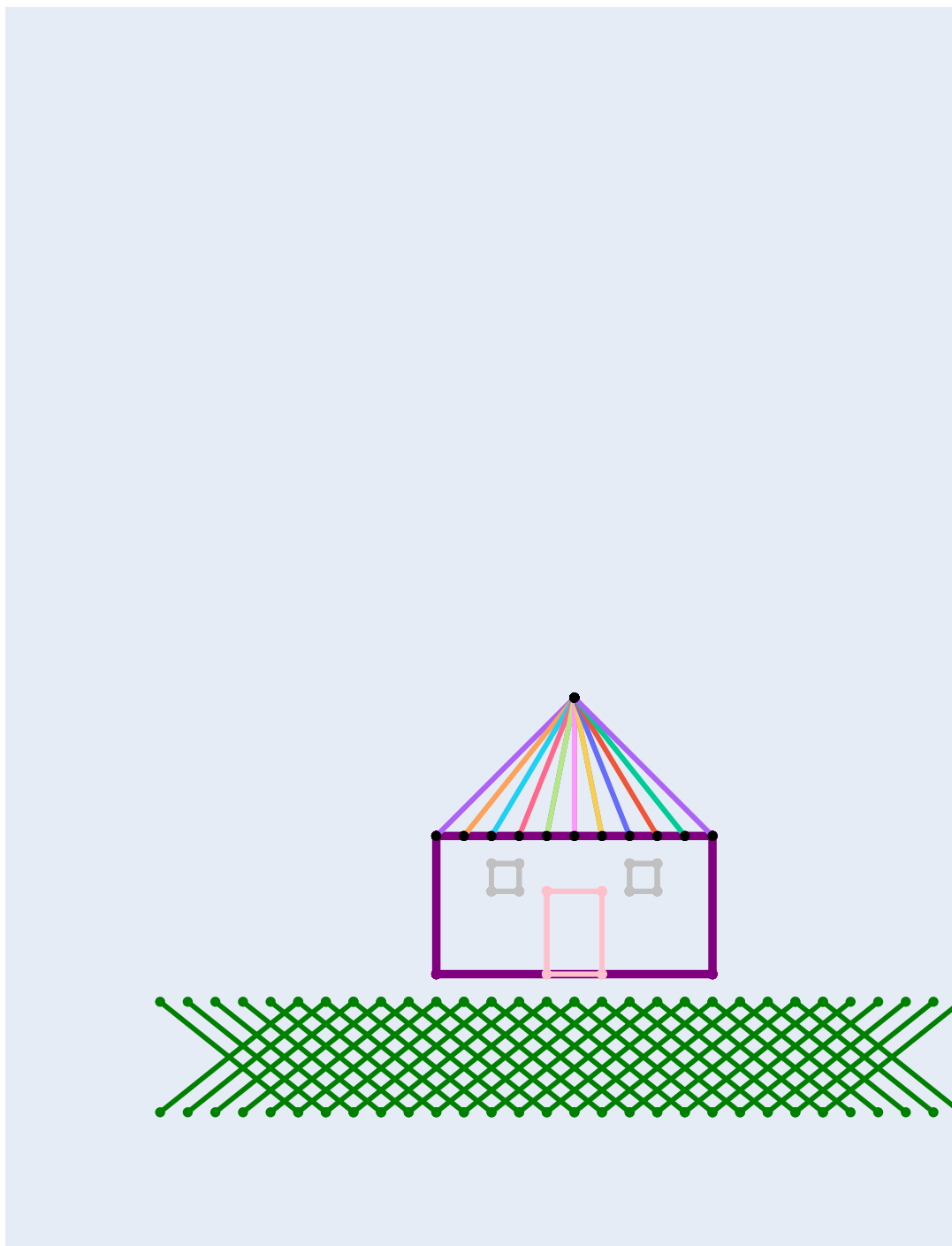
y1 = -15
dx = -5
dy = 4
for x1 in range(-10, +15):
    x2 = x1 + dx
    y2 = y1 + dy
    NiceGr.lineseg(x1=x1, y1=y1, x2=x2, y2=y2, color="green")

# Show the graph
NiceGr.plotAll()
NiceGr.saveImage("frame4.png")
```









In [ ]:

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